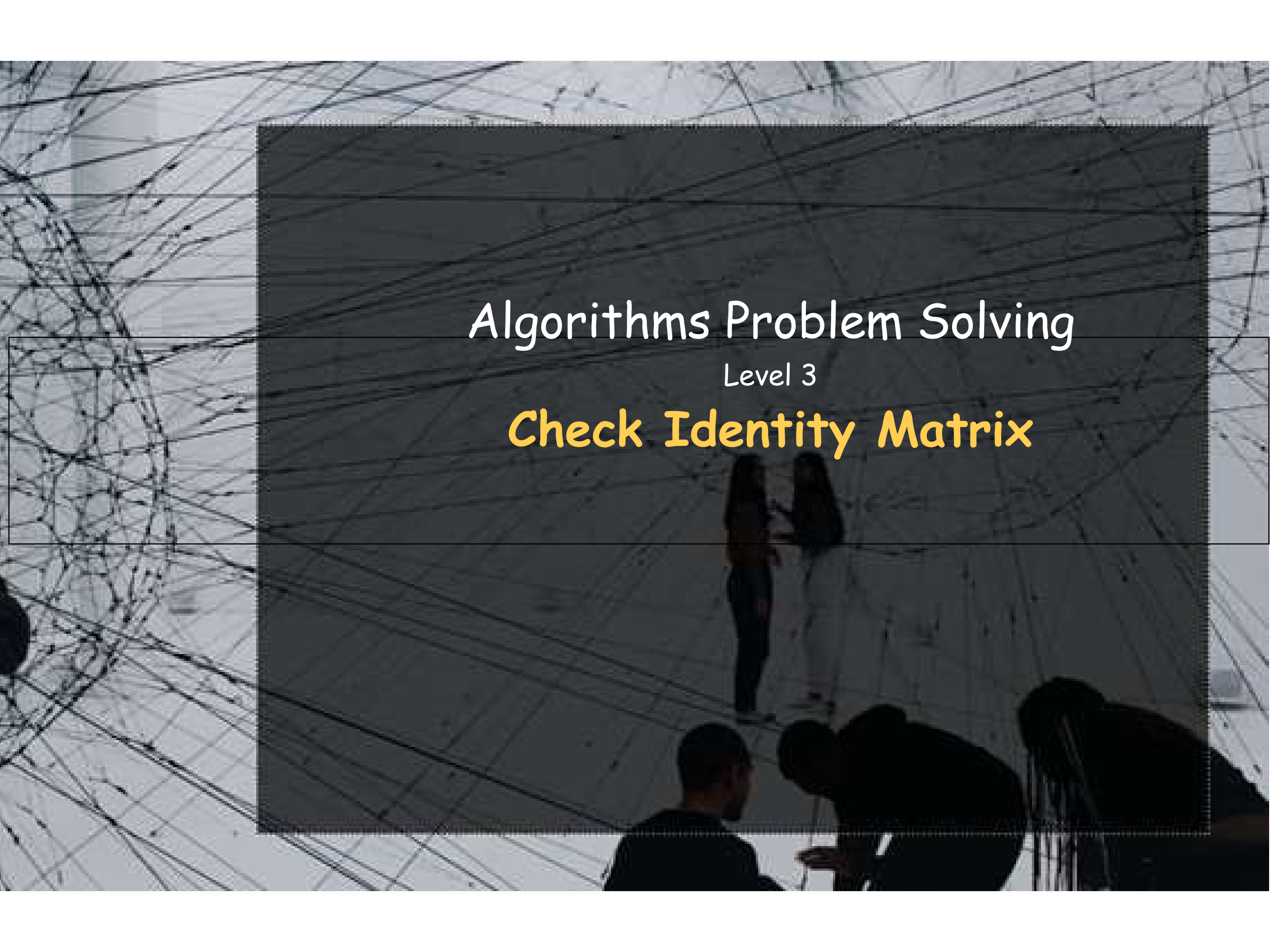


Algorithms Problem Solving

Level 3

Check Identity Matrix



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Definition

Identity Matrix Identification

 Program checks whether a square matrix is an Identity Matrix.

 An identity matrix has `1` on every main diagonal position and `0` at every other position.

Explanation

Main Diagonal Validation

-  For positions where `i == j`, the value must be exactly `1`.

-  These positions form the main diagonal.

Off-Diagonal Validation

-  For positions where `i != j`, the value must be exactly `0`.

-  Every non-diagonal element must therefore be zero.

Mathematical Property

-  The identity matrix acts as the multiplicative identity in matrix multiplication.

-  For a compatible matrix `A`, multiplying by an identity matrix leaves `A` unchanged.

Early Failure Detection

-  If any diagonal or off-diagonal element violates the required condition, the function can immediately return `false`.

Procedures

1. Define `PrintMatrix()`.
2. Define `IsIdentityMatrix()`.
3. Verify that the matrix is square.
4. Check diagonal elements for `1`.
5. Check off-diagonal elements for `0`.
6. Display YES or NO.

💡 Example

matrix1 = { {1,0,0}, {0,1,0}, {0,0,1} };

➡ Result

YES: Matrix is identity.

Note

- ◆ Identity matrices are square matrices.
- ◆ A 3×3 identity matrix contains three `1`s and six `0`s.
- ◆ The identity matrix is also a scalar matrix with scalar value `1`.
- ◆ The identity matrix is also a diagonal matrix.

⊘ Warning

- ◆ Checking only the diagonal elements is not sufficient.
- ◆ A matrix with `1`s on the diagonal but non-zero off-diagonal elements is not an identity matrix.
- ◆ The matrix must have equal numbers of rows and columns.



Comparison

`IsIdentityMatrix()`

- ✓ Requires diagonal elements to equal `1`.
- ✓ Requires every off-diagonal element to equal `0`.
- ✓ Represents the special multiplicative identity matrix.

`IsScalarMatrix()`

- ✓ Requires equal diagonal elements but allows any common scalar value.
- ✓ Requires all off-diagonal elements to equal `0`.
- ✓ Identity matrix is a special case where the scalar value is `1`.

✦✦ Summary

- ✓ Verifies whether a matrix follows the mathematical identity matrix definition.
- ✓ Checks every main diagonal position for `1`.
- ✓ Checks every off-diagonal position for `0`.
- ✓ Demonstrates the special role of the identity matrix in matrix multiplication.

Thank you 😊